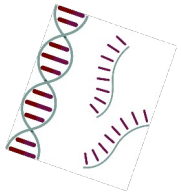
Introduction to Nanopore (ONT) sequencing

ONT mNGS pipeline

Study design



xNA extraction



Library prep



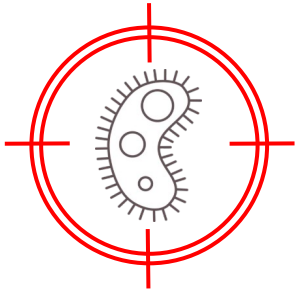
Sequencing



Data analysis



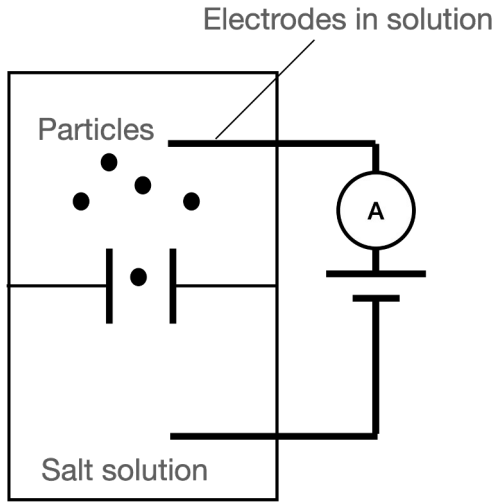
Pathogen ID



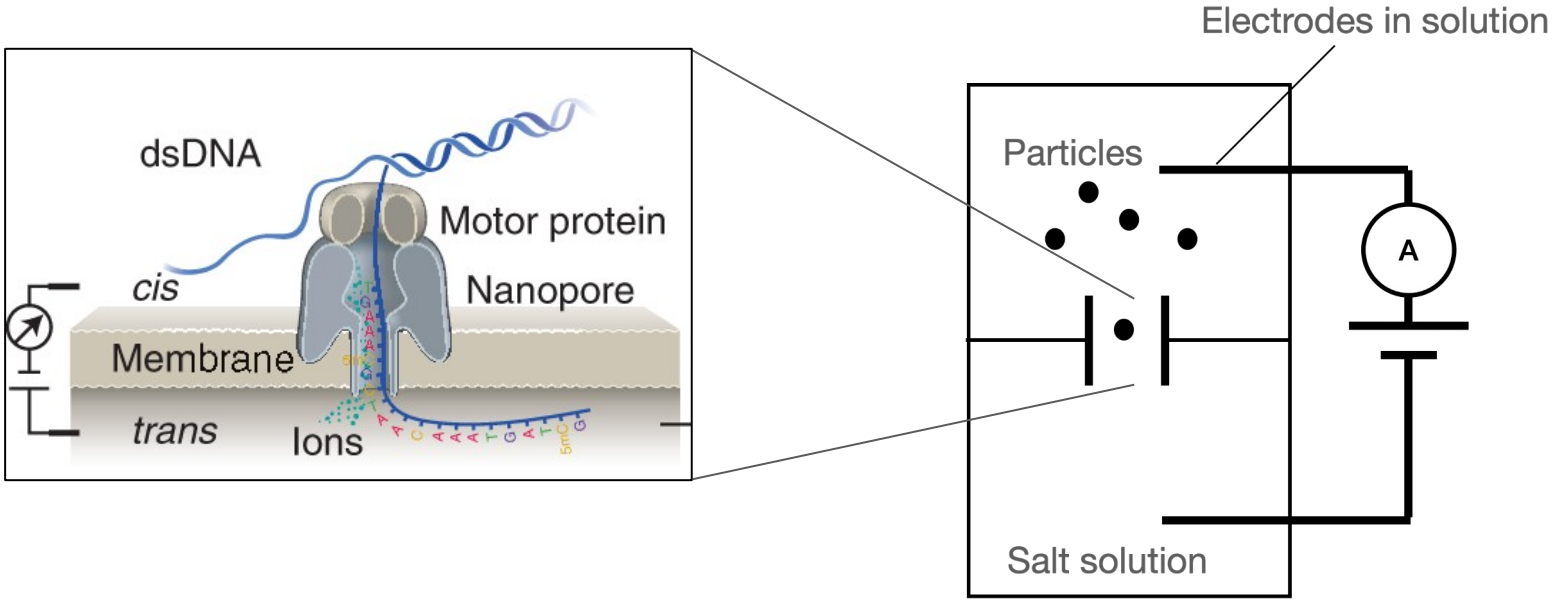
INTRODUCTION

What is nanopore sequencing?

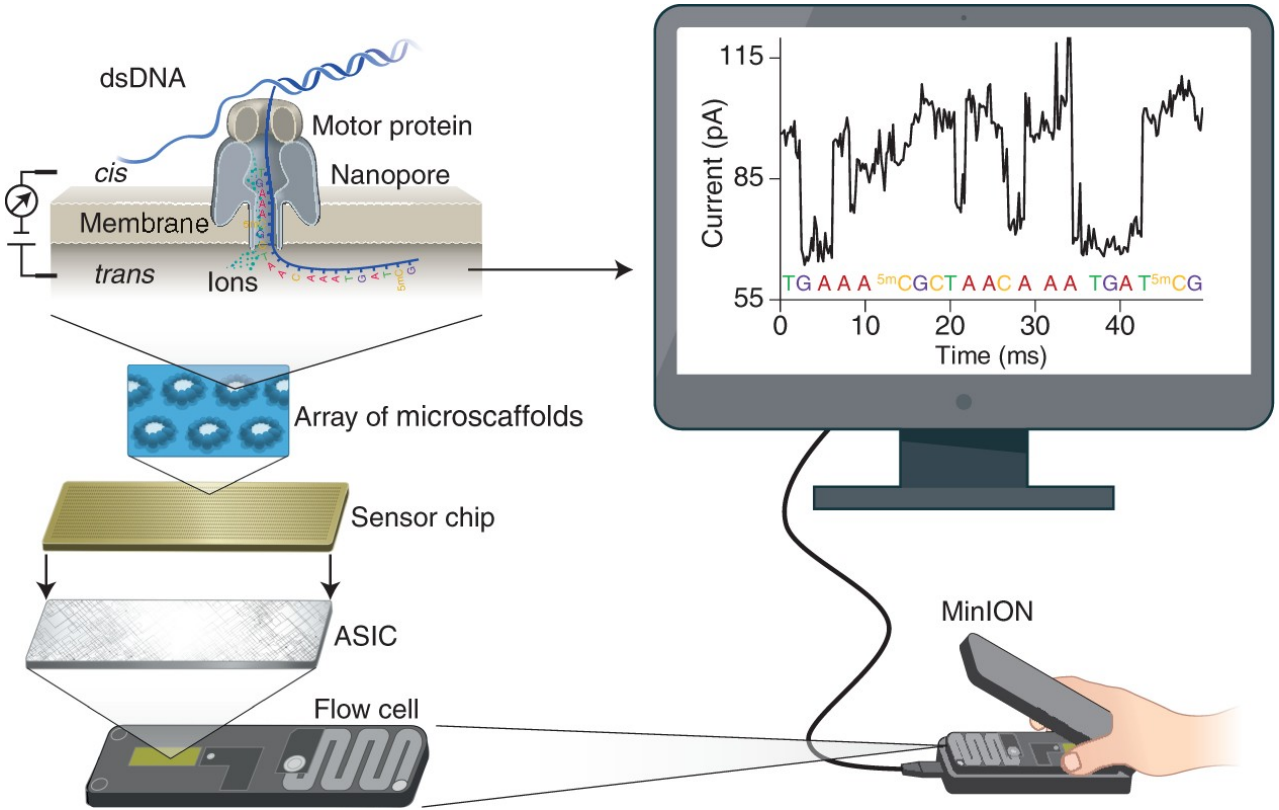
- Use of nano-scale sensors to detect sequence of nucleotides
- Sensors work by detecting changes in electrical current as DNA/RNA particles flow through a hole (“pore”)
- ONT is one of two commercially available nanopore platforms (other is Qitan)



Principles of Oxford Nanopore sequencing



Principles of Oxford Nanopore sequencing

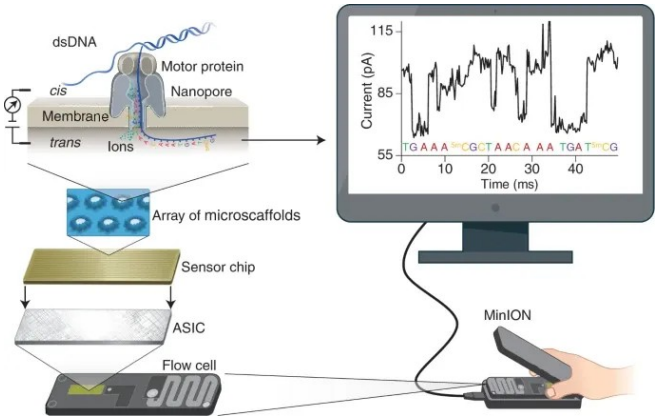


https://www.nature.com/articles/s41587-021-01108-x

Principles of Oxford Nanopore sequencing

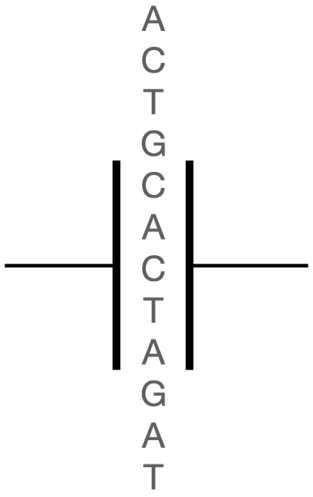
- Biological pores and motor proteins used to guide DNA/RNA across electrical gradient

- Changes in electrical signal during this passage (due to blocking of electrons) detected electronically



Principles of Oxford Nanopore sequencing

- Size of the pore accommodates 5 nucleotides
- DNA/RNA translocation typically happens at ~million bases/second, which is too fast
- Translocation is slowed down to ~1000 bases/second in ONT to allow for discrimination of signals



Different ONT instruments

NANOPORE PRODUCT FAMILY

One core technology, real-time, on-demand

Except for the GridION these are being deprecated











Key	SmidgION	Flongle	MinION	GridION	PromethION
System Price	TBC	Included in $5K Starter Pack	Included in $1K Starter Pack	Included in $50K Starter Pack	Included in $135K Starter Pack
Number of channels	200 channels	128 channels	512 channels	5 x 512 = 2,560*	48 x 3,000* = 144,000
Per flow cell Current Data – Max Data	TBC	1 - 3.3 Gb	17 - 40 Gb	17 - 40 Gb	125 - 311 Gb
Per Device Current Data – Max Data				85 - 200 Gb	3/6 - 20 Tb
Price per Gb Current Data – Max Data	**TBC**	**$90 - $30**	**$30 - $12.5**	**$17.5 - $7.5**	**$5 - $2**

Different ONT instruments 2026



MinION Mk1D



PromethION Series



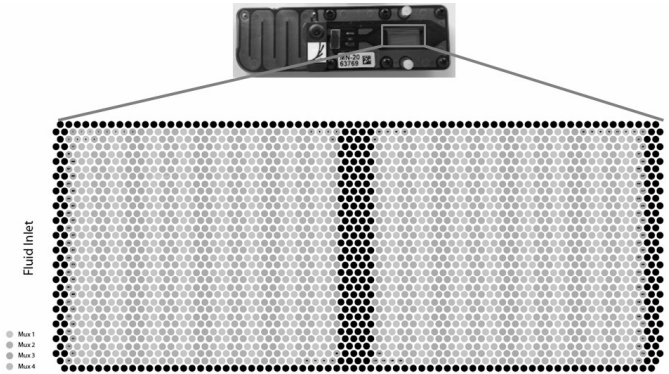
GridION

Instrument	Flow cells at once	Computing	Format
MinION Mk1D	1	External computer	Portable
GridION	5	Built-in	Benchtop
PromethION	2 / 24 / 48*	Model-dependent	Benchtop

*P2 / P24 / P48; P2 Solo needs external compute.

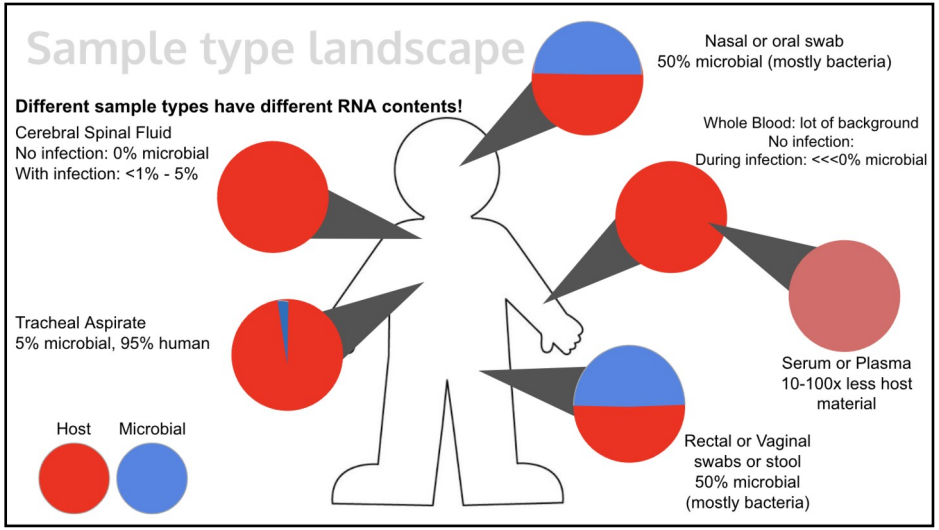
The MinION flow cell

- Each flow cell has 2048 pores
- Typically only ~1200-1600 are actually available on a new flow cell
- Signal from 512 pores can be detected simultaneously



Considerations when doing metagenomics (mNGS)

- DNA or RNA mNGS?
  - Purpose of study
  - Pathogen landscape
- Sample type
  - Total nucleic acid abundance
  - Proportion of host vs. microbe



Considerations when doing Nanopore (ONT) sequencing

- Sample integrity
  - Important to ensure high quality with little to no degradation
  - True for Illumina as well, but even more so for ONT

- Library preparation kit
  - DNA or RNA?
  - PCR or PCR-free?
  - Multiplexing or single sample?

Current kit of choice: Rapid PCR barcoding kit (SQK-RPB114.24)

Next: 07-O_MinION loading and sequencing